MATHS MANTHAN
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Subject: Mathematics **WEEKLY CHAPTER TEST – 06 (QUADRATIC EQUATIONS)** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The ratio of the sum and product of the roots of the quadratic equation $5x^2 - 6x + 21 = 0$ is: [1]
(A) 5 : 21 (B) 2 : 7 (C) 21 : 5 (D) 7 : 2

Q2. If one root of the quadratic equation $x^2 - 4x + 3 = 0$ is 1, then the other root is: [1]
(A) 3 (B) -3 (C) 2 (D) 4

Q3. If the roots of $px^2 + qx + 2 = 0$ are reciprocal of each other, then: [1]
(A) $p = 0$ (B) $p = -2$ (C) $p = 2$ (D) $q = 2$

Q4. If $x = 1$ is a common root of the equations $ax^2 + ax + 3 = 0$ and $x^2 + x + b = 0$, then $ab =$: [1]
(A) 3 (B) 3.5 (C) 6 (D) -3

Q5. **Assertion (A):** The equation $(x + 1)^2 - x^2 = 0$ has degree 1 and has only one real root.
Reason (R): On expanding $(x+1)^2 - x^2 = 0$, the x^2 term cancels leaving the linear equation $2x+1 = 0$. [1]
(A) Both A and R are true and R is correct explanation of A. (B) Both are true, but R is not correct explanation.
(C) A is true, R is false. (D) A is false, R is true.

SECTION B: Short Answer Type Questions (Part I)

Q6. Find the nature of the roots of $3x^2 - 4\sqrt{3}x + 4 = 0$. If they are real, find them. [2]

Q7. An express train takes 1 hour less than a passenger train to travel 132 km between Mysore and Bangalore. If the average speed of the express train is 11 km/h more than that of the passenger train, find the average speeds of both trains. [3]

SECTION B: Short Answer Type Questions (Continued) (Total Sec B = 11 Marks)

Q8. Solve for x: [3]

$$\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}, \quad a, b, x \neq 0,\ x \neq -(a+b)$$

Q9. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed was reduced by 200 km/h and the time of flight increased by 30 minutes. Find the original duration of the flight. [3]

SECTION C: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q10. Case Study: Civil Aviation & Weather Delay Dynamics

A commercial passenger airliner operates on a scheduled domestic sector covering an aerial distance of 1500 km. On a monsoon day, due to adverse weather and runway congestion, the aircraft departed 30 minutes (0.5 hour) later than its scheduled departure time. To reach the destination on time, the pilot increased the cruising speed by 100 km/h over the scheduled speed v km/h.

Based on the above aviation data, answer the following questions:

(i) Write expressions for the scheduled flight time and actual flight time in terms of v. [1]

(ii) Formulate the quadratic equation in v in standard form modeling the time constraint. [1]

(iii) Determine the scheduled cruising speed of the aircraft (v). [2]

OR

Find the scheduled duration of the flight. [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★